

E-Waste & Environmental Management

Activity Worksheet

The Hidden Carbon Footprint of Our Gadgets

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Part A: Carbon Footprint Calculation

Use the carbon footprint calculator provided by your faculty.

Parameter	Your Response
Annual Carbon Footprint	<u>16</u> tCO ₂ e / year
National Average	<u>7</u> tCO ₂ e / year
Global Average	<u>19</u> tCO ₂ e / year
Number of Earths Required (if provided)	<u>3</u>

Which category contributed the most?

☐ Transportation ☐ Home Energy ☒ Food ☐ Shopping & Consumption

☐ Electronics / Digital Lifestyle ☐ Other: _____

Which result surprised you the most?

Food

Part B: My Electronic Lifestyle

1. How many smartphones have you owned in the last 10 years?

☐ 1 ☒ 2-3 ☐ 4-5 ☐ More than 5

2. How often do you replace your smartphone?

☐ Every year ☐ Every 2 years ☐ Every 3-4 years ☒ More than 5 years

3. What do you usually do with your old phone?

☐ Keep ☒ Sell ☐ Exchange ☐ Donate

☐ Recycle ☐ Throw away

4. Have you ever repaired an electronic device instead of replacing it?

☒ Yes ☐ No

If yes: Laptop, T.V

5. Have you ever recycled e-waste through an authorized recycler?

☐ Yes ☒ No

Item: _____

Unused Electronic Items at Home	Number
Mobile phones	<u>1</u>
Chargers	<u>1</u>
Earphones	<u>0</u>
Batteries	<u>0</u>
Cables	<u>0</u>
Other	<u>-</u>

Part C: Think Like an Environmental Engineer

Habit	Higher Carbon Footprint	More E-Waste
Buying a new phone every 2 years	☒	☒
Repairing a laptop	☒	☒
Keeping old phones unused in drawers	☒	☒
Recycling electronic devices	☒	☒
Using devices for 5–6 years	☒	☒

Which stage of an electronic product's life cycle contributes the MOST carbon emissions?

☐ Raw material extraction ☒ Manufacturing ☐ Transportation

☐ Usage ☒ Disposal

Why is extending the life of an electronic device environmentally beneficial?

Repairing and reusing electronic devices lowers carbon emissions, conserves natural resources and reduces e-waste.

Part D: Reflection

Three habits that increase my personal carbon footprint:

- i) Driving vehicles for short duration distances.
- ii) Using electricity unnecessarily.
- iii) Use of electronic devices.

Three habits that increase e-waste generation:

- i) Replacing mobile every year.
- ii) Not repairing old devices throwing it when not useful.
- iii) Unused chargers.

Action to Reduce Carbon Emissions & E-waste	Expected Impact
Use Public transport	Reduces CO ₂ emission
Donate or sell old electronics	Encourages reuse
Avoid unnecessary gadget upgrades	Lowers e-waste
Use LED bulbs	Consumes less electricity.

Discussion Questions (Optional)

1. Which stage of an electronic product's life cycle has the highest carbon footprint and why?

Manufacturing has the highest amount of carbon footprint because it uses large amount of raw materials.

2. How does extending the lifespan of smartphones reduce both carbon emissions and e-waste?

Using smartphones for longer time reduces the need for new phone and decrease e-waste.

3. Why is recycling alone not enough for sustainable e-waste management?

Recycling cannot recover all materials, so reducing, reusing and repair are also important.

4. Explain how the 5Rs contribute to lowering the environmental impact of electronic products.

The 5Rs help save resources reduces pollution, lower emissions and decreases e-waste.

5. As an engineering student, what innovative solution would you propose to reduce the carbon footprint associated with electronic devices?

I would ~~develop~~ develop repairable and modular electronic devices that are easy to upgrade and recycle.

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